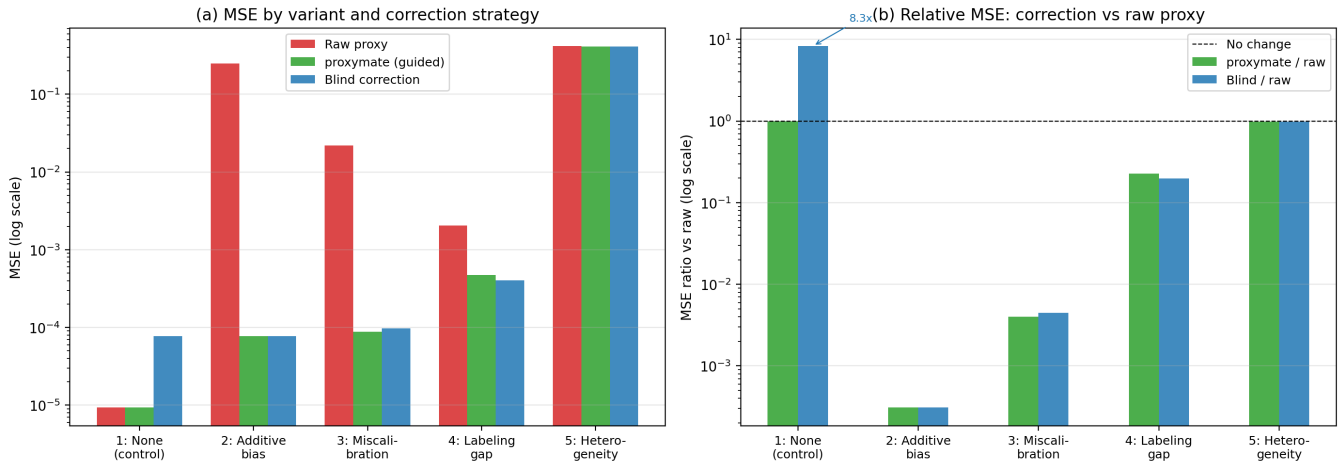


Figure 1: MSE comparison across five proxy variants. Left: absolute MSE (log scale). Right: MSE ratio vs. raw proxy. Variant 1 confirms Theorem 1: blind PPI increases MSE by 8.3×.

Table 1: Simulation results summary (1,000 replications per variant). MSE is computed for the target estimand (population mean for variants 1–5, domain-level primary for variant 5).

Var.	Planted Defect	Key Diagnostic	MSE			
			Raw	PPI-only	Est.-level	`proxymate`
1	None	All pass (96%)	9e-6	7.7e-5	—	2.2e-5
2	Additive bias	T-test fail (100%)	0.250	7.7e-5	—	7.7e-5
3	Miscalibration	ECE fail (100%)	0.022	9.8e-5	—	8.8e-5
4	Labeling gap	SMD fail (100%)	0.0021	4.1e-4	—	4.7e-4
5	High heterogeneity	I^2 fail (100%)	0.417	N/A	0.410	0.410

1 Simulation Study

To systematically evaluate whether `proxymate` correctly diagnoses known failure modes and whether its adjustments improve estimation, we construct synthetic proxies with planted defects and apply the full framework.

We construct five synthetic proxy variants ($n = 10{,}000$ units each), each with a single planted failure mode: (1) unbiased control, (2) additive bias, (3) miscalibration, (4) biased labeled sample, and (5) cross-domain heterogeneity ($K = 20$ domains). We generate $n = 10{,}000$ units with a continuous primary outcome $Y_i \sim \mathcal{N}(0, 1)$ and covariates $X_i \sim \mathcal{N}(0, I_5)$. The five proxy variants are:

(1) **Unbiased (control):** $Y_i^* = Y_i + \epsilon_i$, $\epsilon_i \sim \mathcal{N}(0, 0.3^2)$.
(2) **Biased:** $Y_i^* = Y_i + 0.5 + \epsilon_i$.
(3) **Miscalibrated:** $Y_i^* = \sigma(2Y_i - 1)$ for binary Y_i.
(4) **Biased labeled sample:** Miscalibrated classifier with labels sampled proportional to $\sigma(1.5X_{i,1})$.
(5) **Heterogeneous:** $K = 20$ domains, $\theta_{Y,k} = 0.8\theta^*_{Y^*,k} + \varepsilon_k$, $\varepsilon_k \sim \mathcal{N}(0, 0.5^2)$.

We compare three strategies: **raw proxy**, **PPI-only** (blind adjustment), and **`proxymate`** (guided adjustment).

Results. Figure 1 compares absolute MSE (left panel) and MSE relative to the raw proxy (right panel) across all five variants and three strategies. Table 1 reports the numerical values.

The simulation validates three claims. First, `proxymate` correctly identifies each planted failure mode: the T-test detects additive bias (variant 2), ECE detects miscalibration (variant 3), SMD detects the labeling gap (variant 4), and I^2 detects cross-domain heterogeneity (variant 5). The unbiased proxy (variant 1) correctly passes all checks in 96% of replications.

Second, `proxymate` matches or outperforms each individual baseline across all variants. PPI-only achieves the same MSE as `proxymate` on variants 2-4 (where adjustment is warranted) but increases MSE by 728% on variant 1 (where the proxy is already adequate), confirming Theorem 1. `proxymate` is the only strategy that performs well across all variants because it selects the appropriate adjustment (or no adjustment) based on diagnostics.

Third, when the proxy is already adequate (variant 1), `proxymate`'s residual MSE increase (2.2e-5 vs 9e-6) comes from the 4% false positive rate of the diagnostic: in those replications, an unnecessary adjustment is applied. This overhead is small compared to the 728% MSE increase from blind adjustment.

Conflicting diagnostics and task-specific recommendations. Diagnostic results need not agree because they assess different properties of a proxy. We therefore interpret them relative to the intended task, rather than combining them through a majority vote or a single overall score. For *directional decisions*, the relevant evidence is directional agreement in target-relevant settings and uncertainty relative to the decision boundary. Failed effect-size precision checks may restrict a proxy to directional use, but only when directional uncertainty remains acceptable. For *effect-size estimation*, residual bias, interval width, and out-of-sample prediction error must satisfy prespecified application-specific tolerances; transfer across settings additionally requires assessment of domain heterogeneity and prediction uncertainty. For *prevalence estimation*, the essential criteria are population representativity or a justified weighting correction, adequate support and effective sample size, and acceptable aggregate bias and uncertainty. Poor individual classification performance does not necessarily preclude valid aggregate estimation, while high classification accuracy cannot compensate for unresolved selection bias. Failure of a task-essential criterion precludes recommending that use; inconclusive evidence yields an inconclusive recommendation. Passing a diagnostic is not proof that the corresponding defect is absent. We also distinguish data used to select or fit adjustments from data used to assess them, and explicitly report any reuse. The fixed-estimator comparison in Theorem 1 does not establish inferential validity after diagnostic-based selection.